

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA0716

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4

Total Marks:40

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.
2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.
4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

1.Simple LAN with Two PCs and Switch

Rubrics Answer

Topology:

PC1 → Switch → PC2

IP Configuration

Device	IP Address	Subnet Mask
PC1	192.168.1.1	255.255.255.0
PC2	192.168.1.2	255.255.255.0

Procedure

- Create the topology using one switch and two PCs.
- Configure IP addresses on both PCs.
- Verify connectivity using the **ping** command.
- Capture packets using Wireshark.

Observed Ethernet Frame Fields

Field	Example
Source MAC Address	00:1B:2C:3D:4E:01
Destination MAC Address	00:1B:2C:3D:4E:02
EtherType	0x0800 (IPv4)
FCS	CRC Value (Auto Generated)

Result

The ping was successful and Ethernet frames containing source and destination MAC addresses were observed.

Explanation

The Data Link Layer uses MAC addresses to identify devices within the same LAN. The switch forwards frames based on destination MAC addresses, ensuring correct delivery.

2. Two LANs Connected Through Router

Rubrics Answer

Topology

PC1 → Switch → Router ← Switch ← PC2

IP Configuration

Device	IP Address
PC1	192.168.1.2
Router G0/0	192.168.1.1
Router G0/1	192.168.2.1
PC2	192.168.2.2

Subnet Mask:

255.255.255.0

Default Gateway:

- PC1 → 192.168.1.1
- PC2 → 192.168.2.1

Procedure

- Configure router interfaces.
- Configure PCs.
- Ping between networks.
- Capture ICMP packets.

Observed IPv4 Header Fields

Field	Example
Source IP	192.168.1.2
Destination IP	192.168.2.2
TTL	128
Protocol	1 (ICMP)
Header Length	20 Bytes

Result

Successful communication occurred between both LANs.

Explanation

The Network Layer forwards packets using IP addresses. The router examines destination IP addresses and forwards packets to the appropriate network.

3. ARP Packet Analysis

Rubrics Answer

Topology

Multiple PCs → Switch

Procedure

- Clear ARP cache.
- Ping another host.
- Capture ARP packets.

Observed ARP Packets

ARP Request

Field	Value
Sender IP	192.168.1.1
Sender MAC	00:AA:BB:C C:DD:01
Target IP	192.168.1.2
Target MAC	00:00:00:00:0 0:

Result

ARP Request and Reply packets were successfully captured.

Explanation

ARP converts an IP address into the corresponding MAC address before communication begins, allowing

Ethernet frames to be delivered correctly.

4. ICMP Packet Analysis

Rubrics Answer

Topology

PC1 → Router → PC2

Procedure

- Configure IP addresses.
- Ping destination.
- Capture ICMP packets.

Observed ICMP Fields

Field	Value
Type	8 (Echo Request)
Code	0
Identifier	Random Value
Sequence Number	1
RTT	Approximately 1–5 ms

Echo Reply

Type = 0

Result

Echo Request and Echo Reply messages were successfully exchanged.

Explanation

ICMP is used to verify network connectivity and troubleshoot communication problems. Ping uses ICMP Echo Request and Echo Reply messages to measure reachability and response time.

5.HTTP Client–Server Communication

Rubrics Answer

Topology

PC → Switch → Server

Procedure

- Configure server IP.
- Enable HTTP service.
- Access webpage from client.
- Capture packets.

Captured Communication

TCP Three-Way Handshake

1. SYN
2. SYN-ACK
3. ACK

HTTP Communication

- HTTP GET Request
- HTTP 200 OK Response

Observed Protocols

Protocol	Purpose
TCP	Reliable Connection
HTTP	Web Communication

Result

The webpage was successfully accessed and HTTP communication was captured.

Explanation

TCP establishes a reliable connection through the three-way handshake before transmitting HTTP data. It ensures reliable delivery using acknowledgments, retransmissions, and ordered packet delivery.